DNS Poisoning

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Domain Name System

- Translate host names to IP addresses
  - E.g., www.xyz.com ➔ 74.125.91.103
  - Why is needed?

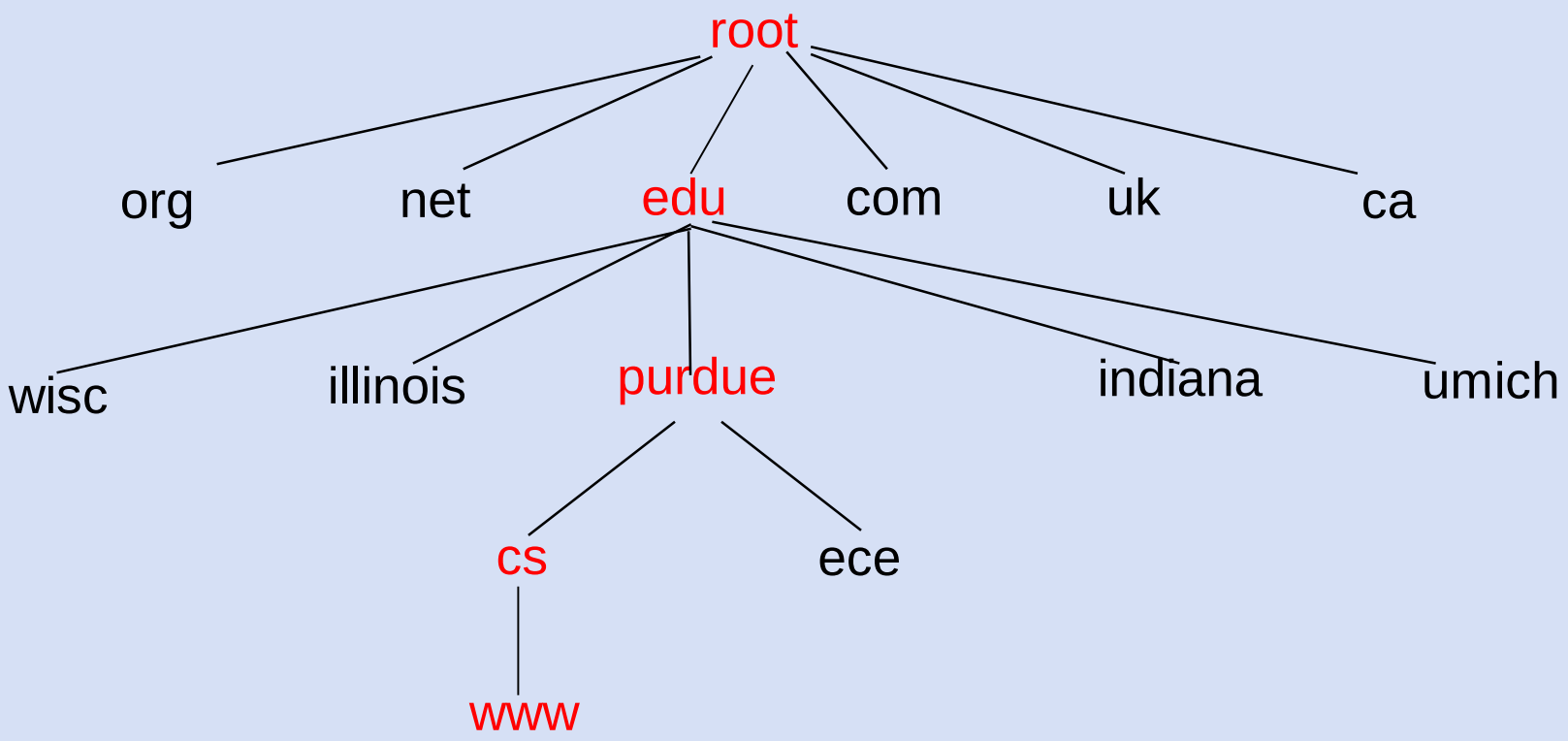
- And back
  - From IP addresses to DNS name

DNS is a Distributed Database

- Information is stored in a distributed way
- Highly dynamic
- Decentralized authority

Domain Name System

- Hierarchical Name Space



Domain Name Servers

- Top-level domain (TLD) servers:
  - responsible for com, org, net, edu, etc, and all top-level country domains, e.g. uk, fr, ca, jp.
  - Network Solutions maintains servers for “.com”
- Authoritative DNS servers:
  - organization’s DNS servers, providing authoritative hostname to IP mappings for organization’s servers.
  - can be maintained by organization or service provider.

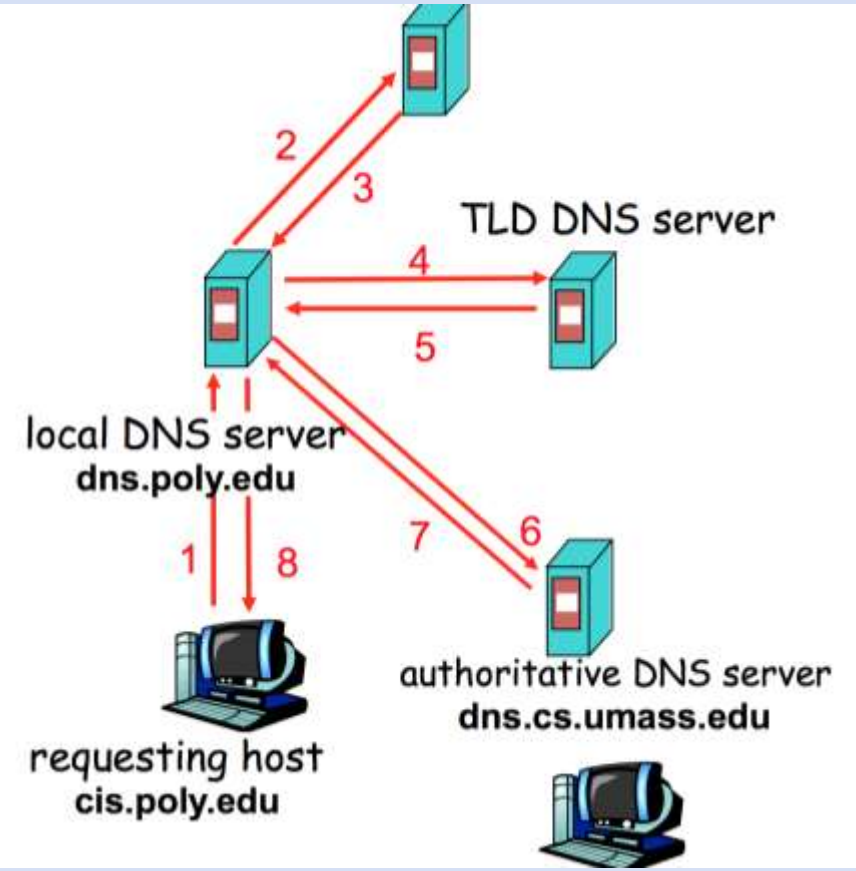
Domain Name Servers - 2

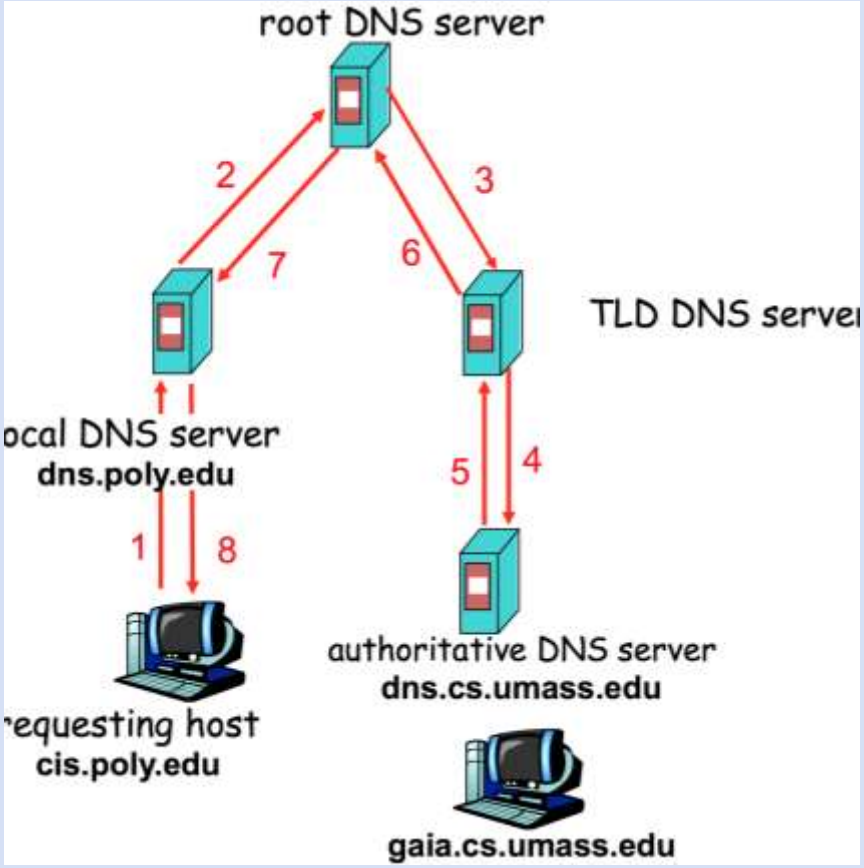
- Local Name Server
  - does not strictly belong to hierarchy
  - each ISP (residential ISP, company, university) has one.

DNS Resolving

- When host makes DNS query, query is sent to its local DNS server.
  - acts as proxy, forwards query into hierarchy.
- Two resolving schemes:
  - Iterative, and
  - Recursive.

DNS Resolving - 2

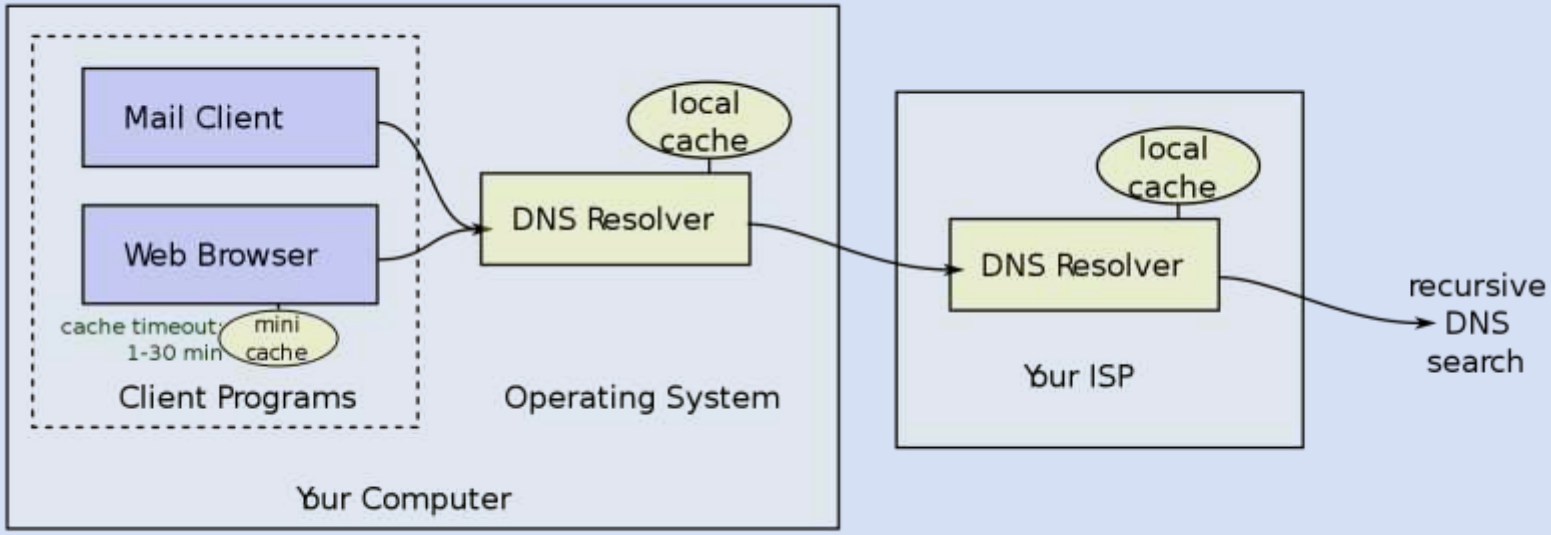




Caching

- DNS responses are cached
  - Quick response for repeated translations
- Negative results are also cached
  - Save time for nonexistent sites, e.g. misspelling
- Cached data periodically times out
  - Each record has a TTL field

Caching - 2


Mail Client
Web Browser
cache timeout:
1-30 min
mini
cache
Client Programs
local
cache
DNS Resolver
Operating System
Your Computer
local
cache
DNS Resolver
Your ISP
recursive
DNS
search

Inherent DNS Vulnerabilities

- Users/hosts typically trust the host-address mapping provided by DNS
  - What bad things can happen with wrong DNS info?
- DNS resolvers trust responses received after sending out queries.
  - How to attack?
- Obvious problem
  - No authentication for DNS responses

User Side Attack - Pharming

- Exploit DNS poisoning attack
  - Change IP addresses to redirect URLs to fraudulent sites
  - Potentially more dangerous than phishing attacks
    - Why?

- DNS poisoning attacks have occurred:
  - January 2005, the domain name for a large New York ISP, Panix, was hijacked to a site in Australia.
  - In November 2004, Google and Amazon users were sent to Med Network Inc., an online pharmacy

DNS Cache Poisoning

- Attacker wants his IP address returned for a DNS query
- When the resolver asks ns1.google.com for www.google.com, the attacker could reply first, with his own IP
- What is supposed to prevent this?
- Transaction ID
  - 16-bit random number
  - The real server knows the number, because it was contained in the query
  - The attacker has to guess

DNS cache poisoning - 2

- Responding before the real nameserver
  - An attacker can guess when a DNS cache entry times out and a query has been sent, and provide a fake response.
  - The fake response will be accepted only when its 16-bit transaction ID matches the query
  - CERT reported in 1997 that BIND uses sequential transaction ID and is easily predicted
    - fixed by using random transaction IDs

DNS cache poisoning (Schuba and Spafford in 1993)

- DNS resource records (see RFC 1034)
  - An “A” record supplies a host IP address
  - A “NS” record supplies name server for domain
- First, guess query ID:
  - Ask (dns.target.com) for www.evil.org
  - Request is sent to dns.evil.org (get quid).
- Second, attack:
  - Ask (dns.target.com) for www.yahoo.com
  - Give responses from “dns.yahoo.com” to our chosen IP.

DNS Poisoning Defenses

- Difficulty to change the protocol
  - Protocol stability (embedded devices)
  - Backward compatibility.
- Long-term
  - Cryptographic protections
    - E.g., DNSSEC, DNSCurve
  - Require changes to both recursive and authority servers
  - A multi-year process
- Short-term
  - Only change the recursive server (local DNS).
  - Easy to adopt

Short-Term Defenses

- Source port randomization
  - Add up to 16 bits of entropy
  - NAT could de-randomize the port
- DNS 0x20 encoding
  - From Georgia tech, CCS 2008
- Tighter logic for accepting responses

Long Term Solution

- DNSSEC:
  - Authenticate responses.
  - Google DNS now is enabled by default.

- Challenges in deployment:
  - Response is large, might no linger fit in single UDP message.
  - Legacy software and machines.